

Inverse of HT Matrix and Chapter 2 Wrap-Up

EGR 445/545

GVSU

Plan for Today

- review
- inverse of an HT matrix
- wrap-up chapter 2
 - Euler angles
 - free vectors
 - computational considerations
- Quiz 1
- LA 6 and/or project time

What does this mean?

Component of $\hat{Z}_B$ along $\hat{X}_A$

$${}^A_B\mathbf{R} = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

Columns of ${}^A_B\mathbf{R}$ are the unit vectors of frame B expressed in frame A components

Unit Vectors and Dot Products

$${}^B\hat{X}_B = ?$$

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$${}^B\hat{X}_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

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$$? = \cancel{{}^A_B\mathbf{R}} \cdot {}^B\hat{X}_B$$

Unit Vectors and Dot Products

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Unit Vectors and Dot Products

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$${}^A\hat{X}_B = {}^A_B\mathbf{R} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Unit Vectors and Dot Products (continued)

$${}^A\hat{X}_B = {}^A_B\mathbf{R} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$


$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = ?$$

$$\begin{bmatrix} a \\ d \\ g \end{bmatrix}$$

$${}^A_B\mathbf{R} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$${}^A\hat{X}_B = ?$$

$$\begin{bmatrix} a \\ d \\ g \end{bmatrix}$$

Dot product Inclusion

$${}^A_B R \cdot {}^B_B \hat{X} = \begin{bmatrix} {}^A_B \hat{X} & {}^A_B \hat{Y} & {}^A_B \hat{Z} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = {}^A_B \hat{X}$$

LA 5 feedback

If you are doing trig manually to find an HT matrix, you are probably doing something wrong.

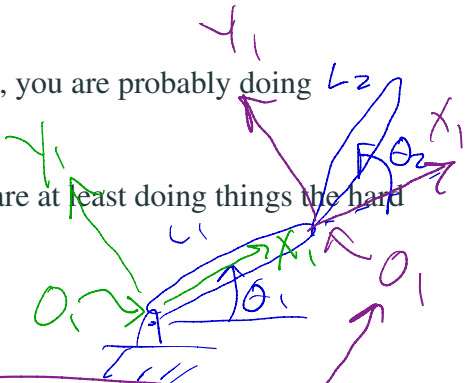
- maybe not getting the wrong answer, but you are at least doing things the hard way
- you might have your origins misplaced

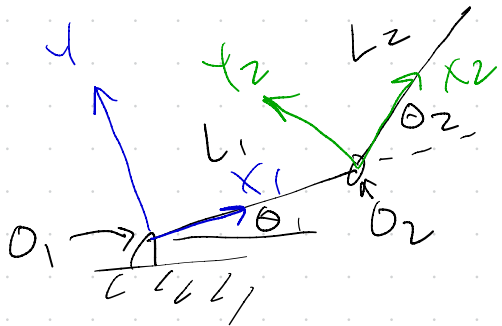
```
P_Tip2 = np.array([8, 0, 0, 1])  
T_12 = robotics.HTz(35, 0, 0, 0)  
T_01 = robotics.HTz(25, 10*cosd(25), 10*sind(25), 0)
```

θ_1

$L_1 = 10$

- Hard way
- also violates DH₅





$${}^0T = HT_z(\theta_1, 0, 0, 0)$$

$${}^1T = HT_z(\theta_2, L_1, 0, 0)$$

$${}^0P_{TIP} = {}^0T @ {}^1T @ {}^2P_{TIP}$$

$${}^2P_{TIP} = \begin{bmatrix} L_2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad {}^1P_{ZORG}$$

Finding the Inverse of an HT Matrix

Goal and Problem Statement

- assuming we know ${}^A_B\mathbf{T}$, how do we find its inverse quickly and cleanly?
- why do we need to do this?
 - what's wrong with doing it numerically?
 - i.e. `numpy.linalg.inv`
- why can't we just use the transpose like with rotation matrices?

HT Inverse Matrix (end goal)

The book throws this out without much explanation:

points from O_B to O_A

$$\begin{matrix} A \\ B \end{matrix} T^{-1} = {}^B_A T = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T \cdot A P_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (\text{eqn. 2.45})$$

- I want you to understand what this means and where it comes from.

$$\begin{aligned} {}^A_B R^T &= {}^A_B R^{-1} = {}^B_A R \\ &\rightarrow -{}^B_A R \cdot A P_{BORG} = - \underbrace{{}^B_A R}_{\text{points from } O_B \text{ to } O_A} \cdot A P_{BORG} \end{aligned}$$

$$A T^{-1} = {}^B A T = \begin{bmatrix} {}^B P & \vdots & {}^B P_{\text{Avg}} \\ \hline 0 & 0 & 0 & \vdots & 1 \end{bmatrix}$$

Goal: Find ${}^B A T$ efficiently from

$$A T$$

$${}^B A T = \begin{bmatrix} A & \vdots & A P_{\text{Avg}} \\ \hline {}^B P & \vdots & {}^B P_{\text{Avg}} \\ \hline 0 & 0 & 0 & \vdots & 1 \end{bmatrix}$$

${}^A_B\mathbf{T}$ vs. ${}^B_A\mathbf{T}$

- what steps does ${}^A_B\mathbf{T}$ capture?
- if ${}^B_A\mathbf{T}$ maps the opposite direction, what steps does it represent?

→ translate A away from B

Letting the notation guide us

$${}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

$${}^A_B \mathbf{T} = \begin{bmatrix} {}^A_B \mathbf{R} & {}^A \mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^A P = {}^A_B \mathbf{T} \cdot {}^B P$$

- what does ${}^B_A \mathbf{T}$ do?
 - what equation does it represent?

Reversing the Mapping

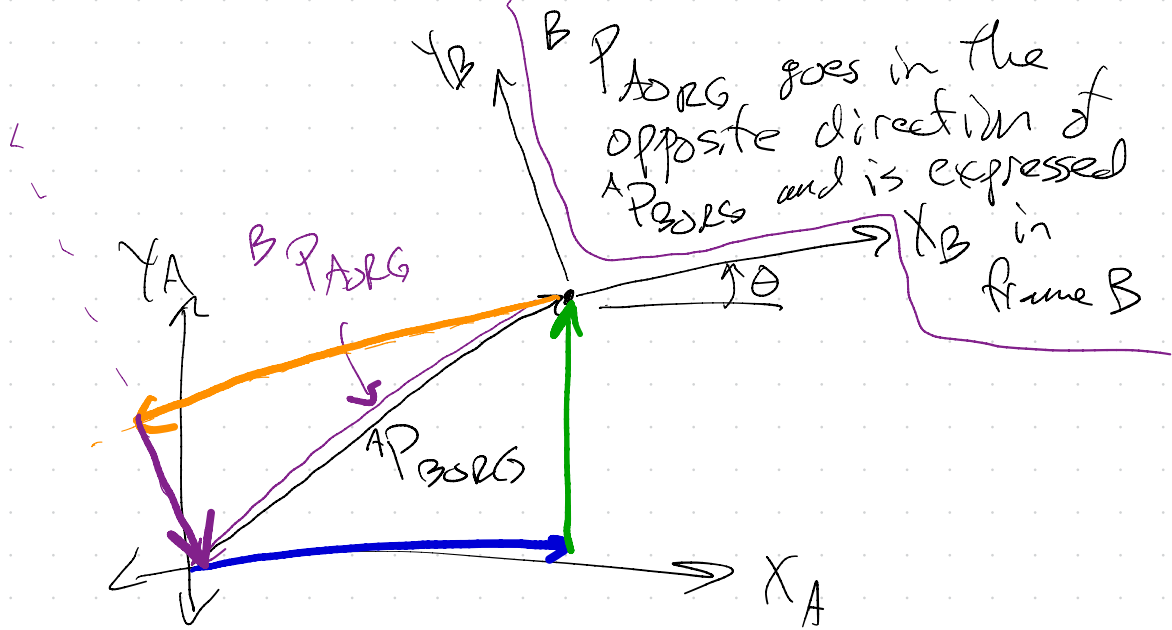
- What does this do? What does an HT matrix combine?

$${}^BP = {}^B_A\mathbf{T} \cdot {}^AP$$

$${}^BP = {}^B_A\mathbf{R} \cdot {}^AP + {}^BP_{AORG}$$

- how do we put it in matrix form?

$$\begin{bmatrix} {}^BP \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B_A\mathbf{R} & {}^BP_{AORG} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^AP \\ 1 \end{bmatrix}$$



Conceptual Question

- conceptually, what is ${}^B P_{AORG}$?

- where does it start and end?

- what frame is it expressed in?

- why is it expressed in that frame?

- what is the relationship between ${}^A P_{BORG}$ and ${}^B P_{AORG}$?

starts @ O_B and ends @ O_A

← frame B

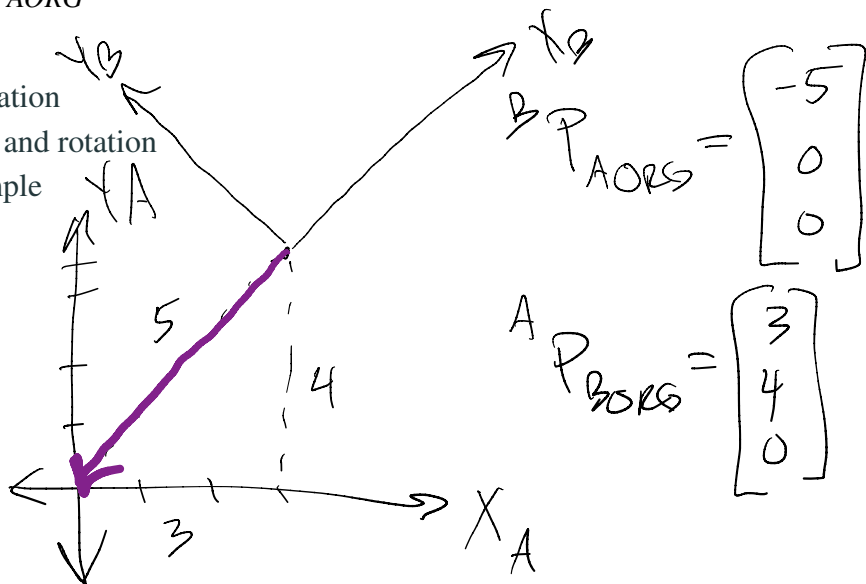
← addition in frame

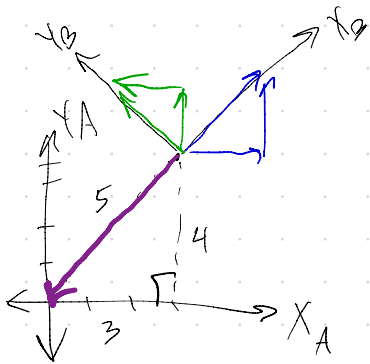
$${}^B P_{AORG} = - \begin{pmatrix} B \\ A \end{pmatrix} R \cdot {}^A P_{BORG}$$

$$\begin{pmatrix} B \\ A \end{pmatrix} R = A \begin{pmatrix} B \\ A \end{pmatrix} R^{-1} = \begin{pmatrix} A \\ B \end{pmatrix} R^T$$

Examples of ${}^B P_{AORG}$

- pure translation
- translation and rotation
- 3/4/5 example





$$A \mathcal{R} = \begin{bmatrix} 0.6 & -0.8 & 0 \\ 0.8 & 0.6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B \mathcal{R} = \begin{bmatrix} 0.6 & 0.8 & 0 \\ -0.8 & 0.6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Pieces of the Reversed Mapping

- how do we find these pieces based on stuff we already know?
 - assuming we have ${}^A_B\mathbf{T}$?

$${}^B_A\mathbf{R} = ?$$

$${}^B\mathbf{P}_{AORG} = ?$$

Finding the Pieces of the Reversed Mapping

$${}^B_A\mathbf{R} = {}^A_B\mathbf{R}^T$$

$${}^B\mathbf{P}_{AORG} = {}^B_A\mathbf{R}(-{}^A\mathbf{P}_{BORG})$$

$${}^B\mathbf{P}_{AORG} = {}^A_B\mathbf{R}^T(-{}^A\mathbf{P}_{BORG})$$

Final Formulation

$$\begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B \mathbf{R}^T & -{}^A_B \mathbf{R}^T \cdot {}^A \mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P \\ 1 \end{bmatrix}$$

HT Inverse Matrix

$${}^B_A\mathbf{T} = \begin{bmatrix} {}^A_B\mathbf{R}^T & -{}^A_B\mathbf{R}^T \cdot {}^A\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (\text{eqn. 2.45})$$

which is the same as

$${}^B_A\mathbf{T} = \begin{bmatrix} {}^B_A\mathbf{R} & {}^B\mathbf{P}_{AORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A\mathbf{P} \\ 1 \end{bmatrix}$$

Given
$$\begin{bmatrix} A^T & B^T \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} A^T P_{\text{ORG}} \\ B^T P_{\text{ORG}} \end{bmatrix}$$

Find
$$B^T = A^T^{-1}$$

extract rows 0, 1, and 2

$$\begin{bmatrix} A^T \\ B^T \end{bmatrix} = \begin{bmatrix} A^T \\ B^T \end{bmatrix} \begin{bmatrix} 0:3 & 0:3 \end{bmatrix}$$

$$B^T P_{\text{ORG}} = -A^T P_{\text{ORG}}$$

$$B^T = A^T^{-1}$$

extract columns 0, 1, and 2

Converting this to Python Code

eye-identity

```
def HTinv(Tin):
```

```
Tout = eye(4)
```

```
R = Tin[0:3,0:3]
```

```
Ri = R.T
```

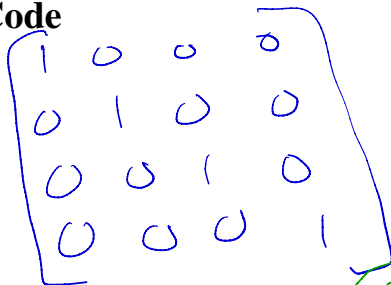
```
Tout[0:3,0:3] = Ri
```

```
P_BorgA = Tin[0:3,3]
```

```
P_AorgB = -1.0*dot(Ri,P_BorgA)
```

```
Tout[0:3,3] = P_AorgB
```

```
return Tout
```



$\begin{bmatrix} R & P \\ A & T \end{bmatrix}$

$\rightarrow$

$\begin{bmatrix} R & P \\ A & T \end{bmatrix}$

$T =$

$\begin{bmatrix} R & P \\ 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

replace zeros

What are Euler angles and why should we care?

- someday you may have to talk to an old, grumpy engineer
- do rotation matrices really have 9 unknowns?

$$\mathbf{R} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

- what constraints are placed on a valid rotation matrix?
 - and where do those constraints come from?

Euler Angles: What do we mean by “arbitrary rotation”?

- are rotation matrices restricted to x , y , or z axes?
- what use would we have for an arbitrary rotation matrix?

Euler Angles

- **big ideas:**

- an *arbitrary* rotation matrix has three unknowns that can be represented using three angles
- an *arbitrary* rotation/orientation can be expressed as the product of three rotation matrices:

$${}^A_B\mathbf{R} = \mathbf{R}(\alpha) \cdot \mathbf{R}(\beta) \cdot \mathbf{R}(\gamma)$$

- we primarily focus on rotations about a single axis
- Euler angles handle the more general case of arbitrary orientation between frames A and B

Free Vectors

What are free vectors and how to we deal with them?

- are some vectors not free?
 - is this a social justice issue?
- is there a whale or dolphin that needs help getting back to the ocean?
- how much do regular vectors cost?
 - do we get free stuff?

Free vs. Line Vectors

- a free vector can be placed anywhere
- the opposite of a free vector is a line vector
- the line of action is important for a line vector
 - you cannot move a line vector without affecting its meaning
- positions and forces are examples of line vectors
- velocities and moments are examples of free vectors

Velocities in different frames

- imagine an AGV in an Amazon warehouse
- the AGV is being observed by two different stations
 - station {B} is rotated and translated from station {A}
 - the velocity of the AGV would have different components in the two frames
 - should the magnitude of the velocity be different in the two frames?

Alternative Velocity Question

$${}^A\mathbf{P} = {}^A_B\mathbf{R} {}^B\mathbf{P} + {}^A\mathbf{P}_{BORG}$$

- velocity is the derivative of position
- taking derivatives in moving reference frames is graduate dynamics
- if ${}^A\mathbf{P}_{BORG}$ is constant, what happens when we take the derivative?

Computational considerations

- what is the fastest way to compute ${}^0\mathbf{P}$?

$${}^0\mathbf{P} = {}^0_1\mathbf{T}_1 {}^1_2\mathbf{T}_2 {}^2_3\mathbf{T}_3 \mathbf{P}$$

Computational considerations: which is faster?

- option 1: Find ${}^0_3\mathbf{T}$ first

$$\begin{aligned} {}^0_3\mathbf{T} &= {}^0_1\mathbf{T} {}^1_2\mathbf{T} {}^2_3\mathbf{T} \\ {}^0\mathbf{P} &= {}^0_3\mathbf{T} {}^3\mathbf{P} \end{aligned}$$

- option 2: Find ${}^2\mathbf{P}$, then find ${}^1\mathbf{P}$, and finally ${}^0\mathbf{P}$

$$\begin{aligned} {}^2\mathbf{P} &= {}^2_3\mathbf{T} {}^3\mathbf{P} \\ {}^1\mathbf{P} &= {}^1_2\mathbf{T} {}^2\mathbf{P} \\ {}^0\mathbf{P} &= {}^0_1\mathbf{T} {}^1\mathbf{P} \end{aligned}$$